G249 Anlysis Report

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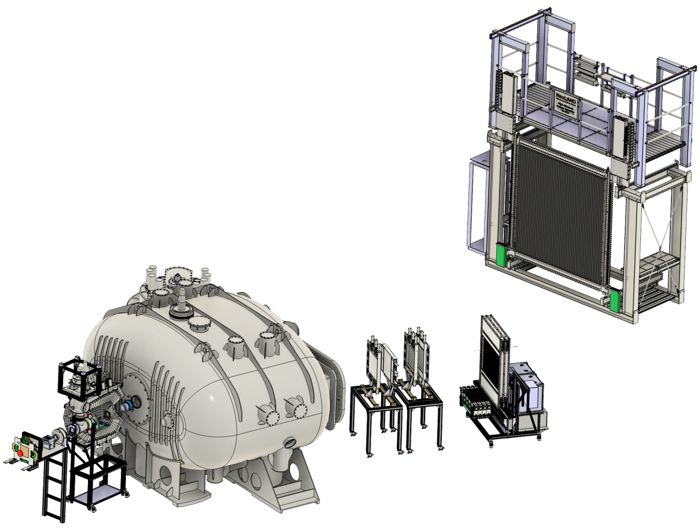

This document is a brief description of the different task performed in the analysis of the G249 experiment. In particular we focus on the work done by the macros of the GitHub repository g249_analysis.

1 Calibrations

These macros check the performance of the calibrations of some detectors (FOOTs and NeuLAND for the moment).

1.1 FOOT

For FOOTs, all the calibrations were done in-flight by R3BROOT (in map2cal and cal2hit classes), so the only correction done after the experiment was charge calibration. The problem is that there is a dependence on the energy with the eta parameter that does not allow to perform a direct conversion between energy and charge (see fig. 1 left).

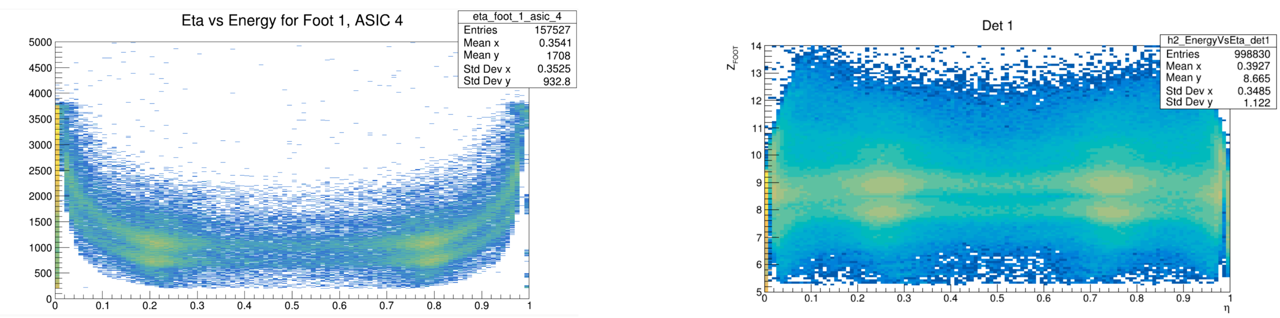

Figure 1: Energy as a function of the eta parameter. Left: With eta correction or charge calibration, only one ASIC. Right: After eta correction and charge calibration for all the ASICs of the detector.

To make the eta correction we perform a high-degree polynomial fit to the $Z = 8$ and $Z = 9$ bands for each ASIC chip and calibrate the energy by making a linear fit (channels, charge) for those bands. This work is done by R3BROOT and, in the macro `check_eta_correction.C`, we plot the charge as a function of eta to show that the former dependence has disappear (see fig. 1 right).

On the other hand, this macro also represent the charge measured in the FOOTs (after all the corrections) against the charge measured in LOS. In fig. 2, we see a clear correlation between both charges. Whereas for charges different from $Z = 8$ and $Z = 9$ we see some saturation, this is not present for our charges of interest.

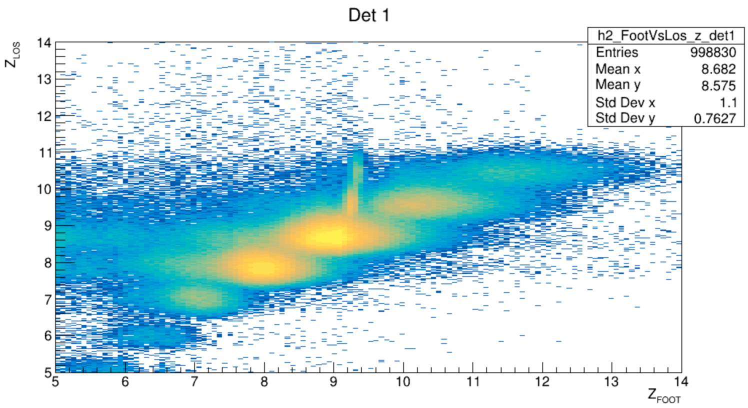

Figure 2: Charge in FOOT vs charge in LOS for the same events

1.2 NeuLAND

For NeuLAND, the calibration consists of calculating the offsets of the time-of-flight of every paddle. To to this, the idea is to align the γ-peak with the nominal value (preliminarily it was set to 51.94 ns, even though it has to be checked).

With the calibration used during the experiment, we got the results shown

2 Simulation

2.1 Neutron decay

Here we simulate the decay of the ^{24}O, produced in (p,2p) reaction, by neutron emission. The idea is to use the generator `R3BNeutronGenerator`. There, we will input the mass and charge of the fragment produced in the de-excitation via neutron emission of the incoming fragment, the kinetic energy per nucleon in the lab frame of the fragment, ~ 650 MeV, the number of emitted neutrons and the energy and width of the resonance that we want to simulate. This generator allows us to do different things:

The first option is to generate resonances. For instance we can try to reproduce the same E_{rel} spectrum as measured in [1]. There, ^{24}O excited states were populated via knock-out reactions from a primary beam of ^{26}F and they measured, in coincidence, one neutron and the outgoing ^{22}O. The result of reconstructing the E_{rel} spectrum (see fig. 3), shows to peaks corresponding to the transitions shown in fig. 4. Although in our experiment we would expect more peaks, since the energy of our beam is higher, we would try to simulate the same spectrum.

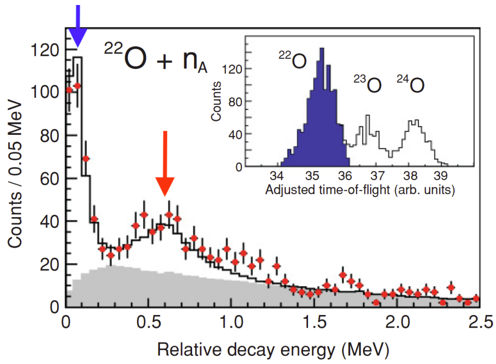

Figure 3: E_{rel} spectrum from [1].

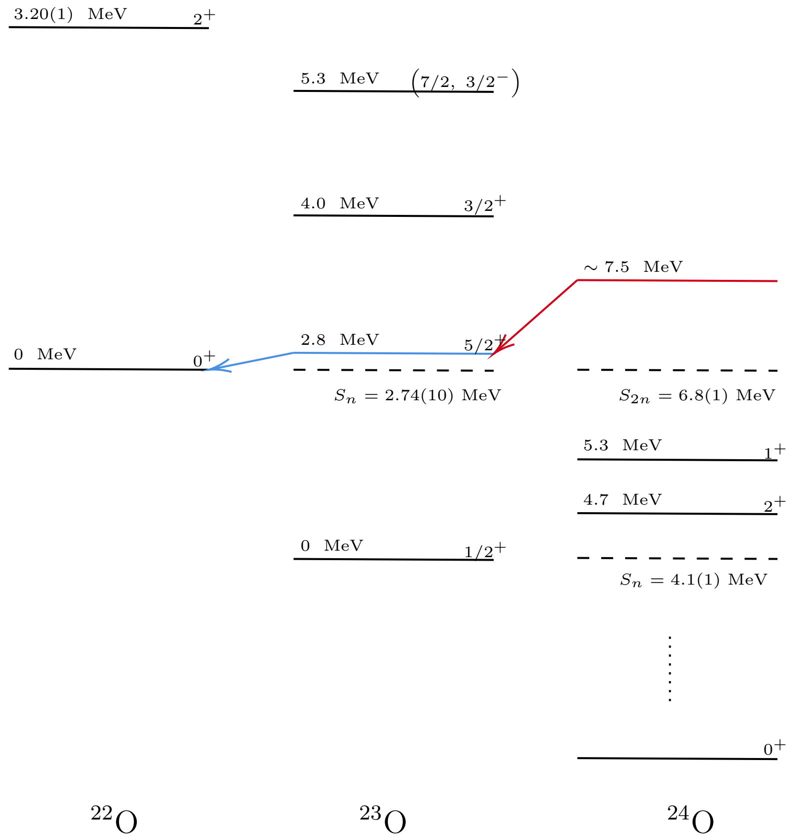

Figure 4: Energy levels and transitions for $^{22,\,23,\,24}O$. In red: Resonant state for the ^{24}O and its transition to the blue resonant state of ^{23}O. In blue: Resonant state of ^{23}O and its transition to the ground state of ^{22}O.

Since in that experiment they were able only to measure one proton during the whole de-excitation process, three possible scenarios are available, $^{24}O^*$ decaying via one neutron emission follow by the emission of a second neutron emission until reaching the ^{22}O. Measuring any of the neutron would result in either blue or read peaks of fig. 3. However, there are still other options, the most obvious one is the simultaneous emission of two neutrons by the $^{24}O^*$, decaying directly to ^{22}O without stopping in the resonant blue level and thus, not contributing to the previous resonant peaks. This case is not going to be explored for the moment.

In order to simulate these resonances we will use the neutron generator to explore the following reactions:

$$^{24}\mathrm{O} \longrightarrow {}^{23}\mathrm{O} + 1n \tag{1}$$
$$^{23}\mathrm{O} \longrightarrow {}^{22}\mathrm{O} + 1n \tag{2}$$

From the results of [1], we know that eq. (1), corresponds to a resonance of ~ 0.6 MeV, while eq. (2), to 0.045(2) MeV. For both cases we can simply consider the kinetic energy of the recoil fragment as the energy of the ^{25}F beam (~ 650 MeV/u). Therefore, running the generator with the following configuration:

```
$ root -l 'NeutronDecayGenerator.C("O24_O23n", 10000, 23, 8, 650.,
    0.6, 0.25)'
$ root -l 'NeutronDecayGenerator.C("O23_O22n", 10000, 22, 8, 650.,
    0.045, 0.02)'
```

will generate two different `.out` files that can be separately read by the simulation class to generate the corresponding events. To run proceed in this way we can simply use the messenger `O24_2n_decay.mac`, this will generate one rootfile per resonance that can be analyzed.

On the other hand, if we want to populate a wider range of energy, instead of a single resonance. We can do as follows:

```
$ root -l 'NeutronDecayGenerator.C("O24_O23n", 10000, 23, 8, 650.,
    eMax, -1)'
```

This creates a spectrum from 0 to `eMax`, which can be usefull to study neutron efficiencies as will be explained in the analysis section.

3 Analysis

3.1

References

[1] C. R. Hoffman, T. Baumann, D. Bazin, J. Brown, G. Christian, D. H. Denby, P. A. DeYoung, J. E. Finck, N. Frank, J. Hinnefeld, S. Mosby, W. A. Peters, W. F. Rogers, A. Schiller, A. Spyrou, M. J. Scott, S. L. Tabor, M. Thoennessen, and P. Voss. Evidence for a doubly magic ^{24}o. *Physics Letters B*, 672:17–21, 2009.